

Criminals use many banks. Now banks can act as one.

ZQUAS lets banks catch scam payments and mule accounts together, at the moment of payment, without pooling customer data. It is built on a GPU governance engine that applies an institution's rules to decisions made by automated systems and AI agents.

750K

COMPLEX EVENTS / SEC

667 ms

500K ENTITIES, 56-POLICY CATALOGUE

TRL 6

SYNTHETIC DATA ONLY

The Problem

- **Scams:**
Scams cost UK victims £576.4 million in 2025, up 19%, across 248,070 cases. The victim sends the money. It lands in a mule account and is quickly moved on. Source: UK Finance, Annual Fraud Report 2026.
- **One piece per bank:**
One mule account often collects from victims at several different banks. Each bank sees only its own customer and its own payment. No bank can see the other banks' pieces.
- **Banks pay:**
Under the UK's mandatory reimbursement rules, in force since October 2024, the sending and the receiving bank split each reimbursement 50:50, up to £85,000 a claim. Banks reimbursed £354.3 million in 2025, 61% of losses. Sources: Payment Systems Regulator; UK Finance, 2026.

The Solution

The receiving bank checks

The bank that holds the receiving account checks the payments coming into it, using its own view of the account and confirmed scam reports shared across the network. It can review a mule account before more money arrives, even when the victim's bank is not in the network.

One report protects every member

A confirmed scam report is shared with every member, so the mule account is flagged for review at every bank in the network. The network only ever adds reviews. It never adds automatic holds.

No customer files leave a bank

No customer files are exchanged, and no raw or directly identifying customer data leaves the institution. Alerts carry proofs that the supervisor can verify. Security is semi-honest. No EU data protection authority has ruled on the status of MPC protocol messages.

Results on synthetic data

In a synthetic world of 1,000,000 adults at the UK scam rate, each bank's own fraud model caught about 1 in 7 scam payments. With the network about 1 in 2 were reviewed, in our model, and about 1 in 3 without the generator's easiest clue. Every limit held. These are not real-world detection rates.

Crypto exchanges and scale

In our model, two crypto exchanges as members caught about a quarter more scam payments. Up to 540 simulated bank installations ran together, and 99 in 100 decisions took under 10 ms, in our test.

The engine underneath

A GPU governance engine applies an institution's rules to decisions made by automated systems and AI agents. 750,000 complex events per second across a full 500,000-entity detection cycle. Financial crime is where it starts.

Position today

TRL 6. Synthetic data only. No production deployments. ZQUAS is taking part in the FCA's Supercharged Sandbox. The FCA does not endorse ZQUAS, its products or its results.

Core Engine

CAPABILITY	DETAIL
Processing speed	667 ms median of three runs for a 500,000-entity full detection cycle against the 56-policy catalogue. 750,000 compl
Policy language	CPL: lexer, parser, semantic analyser, compiler to GPU bytecode
Adjudication pipeline	Triple-stream GPU: Load (H2D), Adjudicate (kernel), Commit (D2H + sign)
Determinism	Byte-identical replay guaranteed. Fixed-point arithmetic where applicable
Entity resolution	GPU-resident high-throughput hash table, GNN risk scoring
Proof generation	Cryptographic proof bundle: verdict hash, Merkle proof, Ed25519 signature
ZK governance proofs	GPU-accelerated zero-knowledge governance proofs

Governance & Safety

- Policy execution gateway: Zero-bypass enforcement point. No code path circumvents governance.
- Authorisation framework: Ephemeral Ed25519 keypairs. Scoped, time-limited, non-reusable action warrants.
- Resource governor: Monotonic per-agent gas ledger. Sharded concurrency. Prevents resource exhaustion.
- Predictive compliance engine: Forward simulation of compliance state. Predicts budget depletion, warrant expiry, and coverage gaps before they occur. Fixed-point deterministic projection.

Detection Intelligence

- Semantic Knowledge Graph: Multi-domain semantic encoding across entity existence, relationships, behaviour, epistemic state, and temporal projections. Contextual reasoning beyond raw transaction patterns.
- Multi-Timeline Evaluation: Same entity evaluated under multiple policy versions simultaneously on GPU. Scenario analysis, impact assessment, and counterfactual replay at compliance speed.
- Cross-Institutional MPC: Federation protocol: ECDH-PSI (X25519) for entity matching, Yao's Garbled Circuits (Free-XOR) for risk comparison, IKNP OT for oblivious transfer. Semi-honest security model. Under 10 seconds per bilateral round. Dual Ed25519 attestation. AES-256-GCM encrypted transport. No raw or directly identifying customer data leaves the institution.
- Financial Crime Network Simulator: GPU-native simulation of complete financial ecosystems with embedded criminal networks across 8 FATF-documented typologies. CBS-calibrated legitimate economy. Falsifiable, reproducible detection measurement. The industry benchmark that does not exist anywhere else.

Continuous Verification

- Adversarial testing framework: 13+ subsystems. GPU-accelerated fuzzing, decision surface cartography, adversarial oracle. Two-gate deployment pipeline (pre-commit + CI/CD).
- GapDetector: Real-time policy coverage mapping against registered regulatory frameworks. Gaps identified automatically, continuously.
- Build Attestation: BLAKE3 + SHA-256 + Ed25519 embedded in binary. Runtime self-verification. Compiler version and git commit recorded. Reproducible builds.

Integration

- Ingest: 256MB shared memory ring buffer. Lock-free atomic indices. JSON/REST, raw binary, GPU-Direct bypass. Configurable backpressure (drop newest, drop oldest, block).
- Export: RAW and CEF formats for SIEM (Splunk, QRadar, Sentinel). Zero-copy flat encoding. Domain-tagged BLAKE3 integrity. Every exported verdict traceable to its proof bundle.

Market Opportunity

- 200B+ euros annual global compliance spending. Growing 15-20% per year. Banks allocate 10-20% of operating budgets to compliance.
- AMLR mid-2027 creates a hard regulatory deadline. Every European bank must re-architect. EU AI Act classifies AML as high-risk with mandatory technical standards from August 2026.
- Cross-institutional detection proven valuable (TMNL demonstrated this over 4 years before encountering unresolved legal and proportionality objections) but centralised architecture failed on privacy. MPC-based alternative has no competitors.

Competitive Landscape

	LEGACY INCUMBENTS	AI-NATIVE CHALLENGERS	ZQUAS
Examples	NICE Actimize, Oracle, SAS	Sardine (\$145M), Hawk (\$56M)	---
Architecture	CPU batch, rules-based	CPU cloud, ML-based	GPU-native, MPC + ZK
Cross-bank detection	No	No	Yes (validated, pilot-ready)
Cryptographic proof	No	No	Yes (Ed25519, Merkle)
Deterministic eval	No	No	Yes (byte-identical)
Real-time at scale	Limited	Partial	667 ms for a 500K-entity cycle, 56-policy catalogue.
Adversarial self-testing	No	No	Continuous (GPU)
Regulator verification	Trust-based	Trust-based	Independent CLI

Traction

- DNB InnovationHub: Submission under review
- FCA Digital Sandbox: Accepted (March 2026). ZQUAS is taking part in the FCA's Supercharged Sandbox. The FCA does not endorse ZQUAS, its products or its results.
- FCA Supercharged Sandbox: Second cohort, one of 21 firms (July 2026). ZQUAS is taking part in the FCA's Supercharged Sandbox. The FCA does not endorse ZQUAS, its products or its results.
- NVIDIA Inception program member
- Benchmark: 667 ms median of three runs for a 500,000-entity full detection cycle against the 56-policy catalogue, 2-second bound enforced in CI. RTX 5090, CUDA 13.1, Linux. 750,000 complex events/sec sustained.
- Cross-institutional detection: under 10 seconds per bilateral round. Correspondent banking wire stripping validated end to end as a named cohort on synthetic data.
- Maturity: TRL 6. Synthetic data only, no production deployments, semi-honest security model.
- Regulatory frameworks: EU AI Act, AMLR, FATF R15, NIST AI RMF, ISO 42001, MAS TRM, GDPR, DORA

Business Model

Land:

Single-bank on-premise deployment. Annual licensing by transaction volume and policy complexity. Bank's data never leaves its infrastructure.

Expand:

Additional business lines, policy domains, jurisdictions within existing banks. Net revenue retention above 100%.

Network:

Cross-institutional MPC activates as the federation grows. Each new bank increases detection for all participants. Switching costs deepen with cryptographic audit chain.

How Cross-Institutional Detection Works

Banks already monitor transactions. Criminals exploit the gaps between banks. ZQUAS adds a federation layer that makes cross-institutional patterns visible without sharing any data. Each bank keeps its existing systems and processes.

What Single-Bank Monitoring Misses

A criminal entity has accounts at three banks. At each bank, the transaction pattern looks normal. Across all three, the pattern is classic money laundering: money enters through one bank, fragments through a second, and reconsolidates through a third. No single bank can see this.

From Transaction to SAR in Six Steps

- 1. Local Monitoring**
Banks monitor transactions using their existing systems. Risk scores per entity based on local patterns. Unchanged.
- 2. Federation Round**
ZQUAS installations execute bilateral rounds (A-B, A-C, B-C). Private Set Intersection reveals shared entities. Secure comparison checks combined risk against threshold. Neither bank sees the other's score. Ed25519 attestation per round.
- 3. Escalation Alert**
If combined risk exceeds threshold, both banks receive an alert: entity ID, escalation flag, attestation proof. No transaction data. No individual scores. Delivered to existing case management via API or file import.
- 4. Enhanced Due Diligence**
Analyst receives the alert in their normal workflow. Reviews their own transactions. The new information: this entity has elevated cross-institutional risk.
- 5. SAR Decision**
Each bank independently decides whether to file with FIU-Nederland. The bank uses only its own data in the SAR. The cross-institutional alert is the trigger, not the content.
- 6. FIU Investigation**
FIU receives separate SARs from multiple banks about the same entity. The complete laundering pattern becomes visible. No single bank could have seen it alone.

Measured on Production Hardware (RTX 5090)

METRIC	RESULT
Single-bank pipeline (500K entities, 56-policy catalogue)	667 ms, median of three
Bilateral federation round (100K entities per party)	3,052 ms (loopback transport); under 10s bound in CI
Sustained throughput	750,000 complex events/sec across the 500K-entity cycle
Typology validated	Correspondent banking wire stripping, end to end (synthetic)
Transport security	AES-256-GCM + Ed25519 on every round
Data shared between banks	No raw or directly identifying customer data leaves the institution

Correspondent banking wire stripping validated end to end as a named cohort on synthetic data across simulated participants. Trade-based over-invoicing, shell company layering and funnel account structuring are designed and in scope, not validated. No raw or directly identifying customer data leaves the institution.

What Each Bank Learns

One new fact per shared entity: whether the combined risk exceeds the agreed threshold. Not the other bank's score. Not their transactions. One bit of information. Enough to trigger investigation, not enough to compromise privacy.

Team

Danny de Gier Founder & Engineer

Compliance Career

18+ years in financial crime compliance at Tier-1 banks and fintechs. Senior roles at RBS, Deutsche Bank, HSBC, and Commerzbank. Fintech compliance leadership at ClearBank, Vivid Money, and CoinMetro. Professional Postgraduate Diploma in Financial Crime Compliance (International Compliance Association / University of Manchester). Has personally built monitoring frameworks, written SARs, sat through regulatory examinations, and managed compliance teams across multiple jurisdictions.

Engineering

Self-taught GPU systems programmer. C++23, CUDA (sm_86/89/100/120), Vulkan. Built the entire ZQUAS engine as a solo developer: policy language compiler, GPU adjudication kernel, MPC protocol implementation, cryptographic attestation pipeline, adversarial testing framework, entity resolution graph, rendering engine. This dual expertise, compliance domain knowledge combined with GPU systems programming, shapes every architectural decision and is extremely rare in the talent market.

Martin van Aalderen Chief Commercial & Corporate Development Officer

Entrepreneur with 20+ years in global financial markets. Former founder and CEO of Object+, a financial software firm operating across Europe and the United States. Career spanning banking, capital markets, and enterprise software at Object+, ABN AMRO, and ING. Leads commercial strategy, partnerships, and corporate development.

Engineering Standards

STANDARD	DETAIL
Language	C++23 with CUDA
Codebase	Approximately 753K source lines excluding third-party code
GPU targets	sm_86, sm_89, sm_100, sm_120
Build hardening	/sdl, /guard:cf, /GS, /Qspectre, /CETCOMPAT, /HIGHENTROPYVA
Maturity	TRL 6. Synthetic data only, no production deployments, semi-honest security
Cryptography	BLAKE3, SHA-256, Ed25519, ZK-optimised hash functions, SipHash-128
Entity system	High-performance entity component system
Serialisation	Custom binary format (64B-aligned, std140-compatible)
Build attestation	BLAKE3 + SHA-256 + Ed25519 embedded, reproducible builds

Regulatory Framework Coverage

EU AI Act

AMLR / 6AMLD

FATF R15

NIST AI RMF

ISO 42001

MAS TRM

GDPR

DORA

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The architecture is ready. The regulatory window is open. Let's talk.

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